API CONTRACT SPECIFICATION

Complete Endpoint Collection & Event Schemas

Real-Time Luxury Auction Engine • Base URL: `http://localhost:8080/api/v1`

v2.0.0 Corporate Baseline
Status: Approved Baseline
24 October 2026

Metadata	Nilai
Dokumen	Complete API Contract Collection & Response Schemas
Proyek	Real-Time Auction & Bidding Engine (Fintech / E-Commerce)
Versi	v2.0.0 (Corporate Baseline)
Status	`Approved Baseline`
Tanggal	2026-10-24
Base URL	`http://localhost:8080/api/v1`

1. Standar Format Respon Global

Respon Sukses (HTTP 200 / 201)

```
{
  "status": "success",
  "data": {}
}
```

Respon Gagal (HTTP 400 / 401 / 403 / 404 / 409 / 500)

```
{
  "status": "error",
  "error": {
    "code": "ERROR_CODE_STRING",
    "message": "Pesan deskripsi kesalahan dalam bahasa Indonesia."
  }
}
```

2. Koleksi Endpoint Lengkap Beserta Seluruh Respon

MODUL 1: AUTENTIKASI DUAL-LOGIN & RBAC

> **1.** `GET /auth/niro/authorize-url`

Frontend (Next.js) memanggil endpoint ini ketika pengguna mengklik tombol *"Login with NIRO"*. Endpoint ini mengembalikan URL redirect otorisasi SSO NIRO beserta state parameter untuk keamanan CSRF.

- **Headers:** *None*
- **Response 200 OK:**

```
{
  "status": "success",
  "data": {
    "redirect_url": "https://auth.niro.internal/oauth2/authorize?client_id=salt_auction_web&response_ty
    "state": "sec_state_981247"
  }
}
```


---

2. `POST /auth/login/niro`

Setelah Frontend membuka URL redirect NIRO, pengguna berhasil login di sistem NIRO dan FE mendapatkan token/kode NIRO. Frontend kemudian mengirimkan token NIRO tersebut ke Backend melalui endpoint ini.

Logika Pemeriksaan Akun di Backend:

- Backend memvalidasi token ke server NIRO dan mengekstrak identitas (`niro_user_id` , `email` , `username`).
- Backend memeriksa tabel `users` berdasarkan `niro_user_id` :
  - **Jika Belum Terdaftar:** Sistem secara otomatis **mendaftarkan akun baru** (`auth_provider = 'NIRO'` , role default `BIDDER`), membuatkan akun dompet (`available_balance = 0` , `locked_balance = 0`), dan menerbitkan token JWT sesi. Respon mengembalikan status `is_new_user: true` (HTTP 201).
  - **Jika Sudah Terdaftar:** Sistem langsung **meloginkan pengguna**, memuat role terkini (`BIDDER` atau `ADMIN` jika sudah dipromosikan), dan menerbitkan token JWT sesi. Respon mengembalikan status `is_new_user: false` (HTTP 200).
- **Headers:** `Content-Type: application/json`
- **Request Body:**


```
{
  "niro_token": "eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9.niro_sample_token ... "
}
```


- **Response 200 OK (Pengguna Sudah Terdaftar — Berhasil Loginkan):**


```
{
  "status": "success",
  "data": {
    "is_new_user": false,
    "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.auction_jwt_token ... ",
    "user": {
      "id": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
      "niro_user_id": "NIRO-USR-99812",
      "username": "budi_kolektor",
      "email": "budi@collector.id",
      "auth_provider": "NIRO",
      "role": "BIDDER"
    },
    "wallet": {
      "available_balance": 150000000.00,
      "locked_balance": 0.00,
      "total_balance": 150000000.00
    }
  }
}
```

- **Response 201 Created (Pengguna Belum Terdaftar — Berhasil Didaftarkan Otomatis):**

```
{
  "status": "success",
  "data": {
    "is_new_user": true,
    "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.auction_jwt_token ... ",
    "user": {
      "id": "7ca12345-5717-4562-b3fc-2c963f66bbbb",
      "niro_user_id": "NIRO-USR-10293",
      "username": "siti_investor",
      "email": "siti@luxury.com",
      "auth_provider": "NIRO",
      "role": "BIDDER"
    },
    "wallet": {
      "available_balance": 0.00,
      "locked_balance": 0.00,
      "total_balance": 0.00
    }
  }
}
```

- **Response 401 Unauthorized (Token NIRO Invalid/Expired):**

```
{
  "status": "error",
  "error": {
    "code": "ERR_INVALID_NIRO_TOKEN",
    "message": "Token otentikasi NIRO tidak valid atau telah kedaluwarsa."
  }
}
```

---

3. `POST /auth/login/email`

Login terisolasi khusus Superadmin platform lelang menggunakan Email dan Password internal (minimal 12 karakter), dilengkapi proteksi penguncian bertingkat (*Progressive Lockout*).

Aturan Kebijakan Gagal Login (Lockout):

- 3x salah pertama: Terkunci **1 menit**.
- 3x salah kedua (total 6x): Terkunci **2 menit** (dobel waktu).
- 3x salah ketiga (total 9x): Terkunci **1 jam (60 menit)**.
- Catatan: Fitur *Forgot Password* dan *Kickout* ditiadakan.

- **Headers:** `Content-Type: application/json`
- **Request Body:**

```
{
  "email": "superadmin@salt-auction.internal",
  "password": "MasterAdminSalt2026!"
}
```

- **Response 200 OK (Login Berhasil):**

```
{
  "status": "success",
  "data": {
    "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.superadmin_jwt ... ",
    "user": {
      "id": "00000000-0000-0000-0000-000000000001",
      "username": "superadmin",
      "email": "superadmin@salt-auction.internal",
      "auth_provider": "INTERNAL",
      "role": "SUPERADMIN"
    }
  }
}
```

- **Response 401 Unauthorized (Kredensial Salah):**

```
{
  "status": "error",
  "error": {
    "code": "ERR_INVALID_CREDENTIALS",
    "message": "Email atau kata sandi salah.",
    "details": {
      "remaining_attempts_before_lock": 2
    }
  }
}
```

- **Response 429 Too Many Requests (Akun Terkunci Sementara - Lockout):**

```
{
  "status": "error",
  "error": {
    "code": "ERR_ACCOUNT_TEMPORARILY_LOCKED",
    "message": "Terlalu banyak percobaan login gagal berturut-turut. Akun Anda dikunci sementara selama
    "details": {
      "retry_after_seconds": 60,
      "lockout_level": 1
    }
  }
}
```

---

3. `GET /admin/users`

Mengambil daftar pengguna login NIRO untuk dikelola oleh Superadmin.

- **Headers:** `Authorization: Bearer <SUPERADMIN_JWT>`
- **Query Params:** `?page=1&limit=10&role=BIDDER`
- **Response 200 OK:**

```
{
  "status": "success",
  "data": {
    "users": [
      {
        "id": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
        "niro_user_id": "NIRO-USR-99812",
        "username": "budi_kolektor",
        "email": "budi@collector.id",
        "auth_provider": "NIRO",
        "role": "BIDDER",
        "created_at": "2026-10-24T10:00:00.000Z"
      }
    ],
    "pagination": {
      "page": 1,
      "limit": 10,
      "total_items": 1,
      "total_pages": 1
    }
  }
}
```


- **Response 403 Forbidden:**


```
{
  "status": "error",
  "error": {
    "code": "ERR_FORBIDDEN_ACCESS",
    "message": "Akses ditolak. Endpoint ini hanya dapat diakses oleh Superadmin."
  }
}
```


---

4. `PATCH /admin/users/{id}/role`

Superadmin mempromosikan pengguna login NIRO menjadi **ADMIN** (Auction Manager).

- **Headers:** `Authorization: Bearer <SUPERADMIN_JWT>`, `Content-Type: application/json`
- **Request Body:**


```
{
  "role": "ADMIN"
}
```


- **Response 200 OK:**

```
{
  "status": "success",
  "data": {
    "id": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
    "username": "budi_kolektor",
    "previous_role": "BIDDER",
    "new_role": "ADMIN",
    "updated_at": "2026-10-24T12:05:00.000Z"
  }
}
```

- **Response 404 Not Found:**

```
{
  "status": "error",
  "error": {
    "code": "ERR_USER_NOT_FOUND",
    "message": "Pengguna dengan ID tersebut tidak ditemukan."
  }
}
```

MODUL 2: DOMPET LIKUIDITAS & TOP-UP SALDO

5. `GET /wallet/me`

Mengambil informasi saldo pengguna saat ini.

- **Headers:** `Authorization: Bearer <JWT_TOKEN>`
- **Response 200 OK:**

```
{
  "status": "success",
  "data": {
    "user_id": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
    "available_balance": 25000000.00,
    "locked_balance": 75000000.00,
    "total_balance": 100000000.00
  }
}
```

---

6. `POST /wallet/topup`

Menambahkan saldo bebas (*Available Balance*) untuk modal bidding (*Role BIDDER Required*). Sesuai aturan bisnis **US-001**, **hanya Bidder yang dapat melakukan top-up**, dan nominal minimal pengisian adalah **Rp 100.000**.

- **Headers:** `Authorization: Bearer <BIDDER_JWT>`, `Content-Type: application/json`
- **Request Body:**

```
{
  "amount": 50000000.00
}
```

- **Response 200 OK:**

```
{
  "status": "success",
  "data": {
    "topup_amount": 50000000.00,
    "previous_available_balance": 25000000.00,
    "new_available_balance": 75000000.00,
    "locked_balance": 75000000.00,
    "total_balance": 150000000.00
  }
}
```

- **Response 400 Bad Request (Nominal di Bawah Minimal Rp 100.000):**

```
{
  "status": "error",
  "error": {
    "code": "ERR_MINIMUM_TOPUP_AMOUNT",
    "message": "Minimum top up is IDR 100,000.",
    "details": {
      "minimum_required": 100000.00
    }
  }
}
```

- **Response 403 Forbidden (Bukan Bidder):**

```
{
  "status": "error",
  "error": {
    "code": "ERR_FORBIDDEN_ROLE",
    "message": "Akses ditolak. Fitur dompet dan top-up hanya dapat diakses oleh peran Bidder."
  }
}
```

```
"locked_balance": 75000000.00,
"total_balance": 150000000.00
```

}}

```
```

- **Response 400 Bad Request:**

```
{
  "status": "error",
  "error": {
    "code": "ERR_INVALID_TOPUP_AMOUNT",
    "message": "Nominal top-up harus berupa angka positif lebih dari nol."
  }
}
```

MODUL 3: KATALOG LELANG BARANG MEWAH

7. `GET /auctions/active`

Mengambil daftar snapshot seluruh lot barang bernilai tinggi yang sedang aktif di pasar, mendukung pagination. Gambar aset di-return dalam format **Base64 Compressed String** (hasil decode dari binary BLOB di DB).

- **Query Params:** `?page=1&limit=10&category=LUXURY_WATCHES`
- **Response 200 OK:**


```
{
  "status": "success",
  "data": {
    "auctions": [
      {
        "id": "e0b9b3e1-872b-42e7-91a6-f7615cb38421",
        "title": "Cosmograph Chronograph Steel 2023",
        "category": "LUXURY_WATCHES",
        "condition_grade": "MINT",
        "authenticity_guarantee": "Full Set 2023, Box & Papers",
        "image_base64_compressed": " ... ",
        "model_type": "CAPPED",
        "status": "ACTIVE",
        "starting_price": 350000000.00,
        "current_price": 380000000.00,
        "min_increment": 5000000.00,
        "allowed_increment_steps": [5000000.00, 10000000.00, 25000000.00],
        "valuation_ceiling": 500000000.00,
        "highest_bidder_username": "andi_collector",
        "end_time": "2026-10-25T12:00:00.000Z"
      }
    ],
    "pagination": {
      "page": 1,
      "limit": 10,
      "total_items": 1,
      "total_pages": 1
    }
  }
}
```


---

8. `GET /auctions/{id}`

Mengambil detail spesifikasi satu lot barang bernilai tinggi beserta riwayat penawaran terakhir.

- **Response 200 OK:**

```
{
  "status": "success",
  "data": {
    "lot": {
      "id": "e0b9b3e1-872b-42e7-91a6-f7615cb38421",
      "title": "Cosmograph Chronograph Steel 2023",
      "category": "LUXURY_WATCHES",
      "condition_grade": "MINT",
      "authenticity_guarantee": "Full Set 2023, Box & Papers",
      "description": "White dial, pristine condition, unworn.",
      "image_base64_compressed": " ... ",
      "model_type": "CAPPED",
      "status": "ACTIVE",
      "starting_price": 350000000.00,
      "current_price": 380000000.00,
      "allowed_increment_steps": [5000000.00, 10000000.00, 25000000.00],
      "valuation_ceiling": 500000000.00,
      "highest_bidder_id": "8bb30e92-3a52-47d1-ba76-218f2d5f22e1",
      "highest_bidder_username": "andi_collector",
      "start_time": "2026-10-24T12:00:00.000Z",
      "end_time": "2026-10-25T12:00:00.000Z"
    },
    "recent_bids": [
      {
        "bid_id": "7fa12345-872b-42e7-91a6-f7615cb38421",
        "bidder_username": "andi_collector",
        "amount": 380000000.00,
        "created_at": "2026-10-24T12:15:30.000Z"
      }
    ]
  }
}
```


- **Response 404 Not Found:**


```
{
  "status": "error",
  "error": {
    "code": "ERR_AUCTION_NOT_FOUND",
    "message": "Lot lelang dengan ID tersebut tidak ditemukan."
  }
}
```

MODUL 4: PENAWARAN INKREMENTAL & SALDO TERTAHAN (HOT-PATH)

9. `POST /auctions/{id}/bids`

Mengajukan tawaran lelang dengan memilih tombol inkremental resmi (atau menawar langsung nilai ceiling). Sistem secara otomatis mengevaluasi:

- Jika `current_price + increment_amount ≥ valuation_ceiling`, lelang **LANGSUNG LAKU (*Instant Buyout*)** seketika tanpa perlu flag manual.
- **Headers:** `Authorization: Bearer <BIDDER_JWT>`, `Content-Type: application/json`
- **Request Body:**

```
{
  "increment_amount": 10000000.00
}
```

- **Response 200 OK (Tawaran Sukses & Memimpin):**

```
{
  "status": "success",
  "data": {
    "auction_id": "e0b9b3e1-872b-42e7-91a6-f7615cb38421",
    "new_current_price": 390000000.00,
    "bid_status": "WINNING",
    "wallet_state": {
      "locked_collateral_for_this_bid": 390000000.00,
      "remaining_available_balance": 10000000.00,
      "total_balance": 400000000.00
    },
    "time_extended": false,
    "new_end_time": "2026-10-25T12:00:00.000Z",
    "buyout_triggered": false
  }
}
```

- **Response 200 OK (Mencapai/Melebihi Ceiling — Instant Buyout Terpicu Otomatis):**

```
{
  "status": "success",
  "data": {
    "auction_id": "e0b9b3e1-872b-42e7-91a6-f7615cb38421",
    "new_current_price": 500000000.00,
    "bid_status": "WINNING",
    "wallet_state": {
      "locked_collateral_for_this_bid": 500000000.00,
      "remaining_available_balance": 50000000.00,
      "total_balance": 550000000.00
    },
    "time_extended": false,
    "buyout_triggered": true,
    "auction_status": "CLOSED_AWARDED"
  }
}
```

- **Response 400 Bad Request (Saldo Kurang):**

```
{
  "status": "error",
  "error": {
    "code": "ERR_INSUFFICIENT_AVAILABLE_BALANCE",
    "message": "Saldo bebas Anda (Rp 25.000.000) tidak mencukupi untuk tawaran sebesar Rp 390.000.000. S
  }
}
```

- **Response 400 Bad Request (Lelang Sudah Ditutup):**

```
{
  "status": "error",
  "error": {
    "code": "ERR_AUCTION_NOT_ACTIVE",
    "message": "Lelang ini telah berakhir atau telah dimenangkan via Buyout."
  }
}
```

MODUL 5: BOT PENAWARAN OTOMATIS (AUTOMATED PROXY BIDDING)

10. `POST /auctions/{id}/proxy-bid`

Mendaftarkan atau memperbarui batas maksimal bot proxy bidding.

- **Headers:** `Authorization: Bearer <JWT_TOKEN>`, `Content-Type: application/json`
- **Request Body:**

```
{
  "max_limit": 450000000.00
}
```

- **Response 200 OK:**

```
{
  "status": "success",
  "data": {
    "auction_id": "e0b9b3e1-872b-42e7-91a6-f7615cb38421",
    "proxy_id": "b1827461-872b-42e7-91a6-f7615cb38421",
    "max_limit": 450000000.00,
    "is_active": true,
    "current_auction_price": 395000000.00,
    "your_current_bid": 395000000.00,
    "status": "PROXY_ACTIVE_AND_LEADING"
  }
}
```

- **Response 400 Bad Request (Limit Lebih Kecil dari Harga Saat Ini):**

```
{
  "status": "error",
  "error": {
    "code": "ERR_PROXY_LIMIT_TOO_LOW",
    "message": "Batas limit proxy harus lebih besar dari penawaran saat ini (Rp 390.000.000)."
  }
}
```

---

11. `DELETE /auctions/{id}/proxy-bid`

Membatalkan / menonaktifkan aturan bot proxy bidding aktif.

- **Headers:** `Authorization: Bearer <JWT_TOKEN>`
- **Response 200 OK:**

```
{
  "status": "success",
  "data": {
    "auction_id": "e0b9b3e1-872b-42e7-91a6-f7615cb38421",
    "is_active": false
  },
  "message": "Aturan proxy bidding berhasil dinonaktifkan."
}
```

MODUL 6: MANAJEMEN LOT & KURASI ADMIN

12. `POST /admin/auctions`

Admin / Auction Manager membuat listing lelang barang bernilai tinggi baru.

- Endpoint ini menerima payload dalam format `multipart/form-data` karena menyertakan unggahan binary berkas gambar aset (`image`).
- Gambar binary yang diunggah langsung disimpan di PostgreSQL sebagai **BLOB (`BYTEA`)** dan dikompresi menjadi format Base64 saat dibaca kembali.
- Pengaturan perpanjangan waktu anti-sniping (`anti_sniping_extension_minutes`) dikonfigurasi langsung pada endpoint ini.

- **Headers:** `Authorization: Bearer <ADMIN_JWT>` , `Content-Type: multipart/form-data`
- **Form Data Fields:**
  - `title` (text): `"Togo Leather Handbag 25 Gold Hardware"`
  - `category` (text): `"DESIGNER_BAGS"`
  - `condition_grade` (text): `"MINT"`
  - `authenticity_guarantee` (text): `"Store Receipt 2024, Original Orange Box, Dustbag, CITES cert"`
  - `description` (text): `"Store fresh condition, highly sought-after collector investment piece."`
  - `image` (binary file): `[File binary gambar JPEG/PNG/WEBP]`
  - `min_price` (number): `250000000.00`
  - `allowed_increment_steps` (json string): `"[5000000, 10000000, 20000000]"`
  - `max_price` (number, opsional): `400000000.00`
  - `window_duration_seconds` (number): `86400`
  - `anti_sniping_extension_minutes` (number, opsional): `2`
- **Response 201 Created:**

```
{
  "status": "success",
  "data": {
    "id": "c1f76d22-192a-4c28-8742-1928374aef12",
    "title": "Togo Leather Handbag 25 Gold Hardware",
    "category": "DESIGNER_BAGS",
    "condition_grade": "MINT",
    "authenticity_guarantee": "Store Receipt 2024, Original Orange Box, Dustbag, CITES cert",
    "image_base64_compressed": " ... ",
    "model_type": "CAPPED",
    "status": "ACTIVE",
    "current_price": 250000000.00,
    "valuation_ceiling": 400000000.00,
    "anti_sniping_extension_minutes": 2,
    "start_time": "2026-10-24T12:30:00.000Z",
    "end_time": "2026-10-25T12:30:00.000Z"
  }
}
```


- **Response 400 Bad Request:**


```
{
  "status": "error",
  "error": {
    "code": "ERR_INVALID_AUCTION_PARAMS",
    "message": "Harga maksimal harus bernilai lebih besar daripada harga minimal, dan file gambar wajib
  }
}
```


---

3. Spesifikasi Protokol WebSocket Lengkap

- **URL Koneksi:** `ws://localhost:8080/api/v1/ws/auctions`
- **Handshake Token Query:** `?token=<JWT_TOKEN>` (opsional jika hanya menonton).

3.1 Format Frame Masuk (Inbound Frame dari Client ke Server)


```
{
  "action": "SUBSCRIBE_LOT",
  "lot_id": "e0b9b3e1-872b-42e7-91a6-f7615cb38421"
}
```

3.2 Format Frame Siaran Lengkap (Outbound Events dari Server ke Client)

1. Event `PRICE_UPDATED` (Penawaran Baru Sah)

```
{
  "event": "PRICE_UPDATED",
  "data": {
    "lot_id": "e0b9b3e1-872b-42e7-91a6-f7615cb38421",
    "current_price": 390000000.00,
    "highest_bidder_id": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
    "highest_bidder_username": "budi_kolektor",
    "increment_added": 10000000.00,
    "total_bids_count": 28
  }
}
```

2. Event `TIME_EXTENDED` (Anti-Sniping Aktif)

```
{
  "event": "TIME_EXTENDED",
  "data": {
    "lot_id": "e0b9b3e1-872b-42e7-91a6-f7615cb38421",
    "extension_seconds": 30,
    "new_end_time": "2026-10-25T12:00:30.000Z",
    "reason": "ANTI_SNIPING_TRIGGERED"
  }
}
```

3. Event `AUCTION_BUYOUT_CLOSED` (Harga Maksimal Tersentuh)

```
{
  "event": "AUCTION_BUYOUT_CLOSED",
  "data": {
    "lot_id": "e0b9b3e1-872b-42e7-91a6-f7615cb38421",
    "winner_id": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
    "winner_username": "budi_kolektor",
    "buyout_price": 500000000.00,
    "closed_reason": "VALUATION_CEILING_REACHED"
  }
}
```

4. Event `AUCTION_CONCLUDED` (Selesai Waktu & Settle)

```
{
  "event": "AUCTION_CONCLUDED",
  "data": {
    "lot_id": "c1f76d22-192a-4c28-8742-1928374aef12",
    "status": "SETTLED",
    "winner_id": "8bb30e92-3a52-47d1-ba76-218f2d5f22e1",
    "winner_username": "andi_collector",
    "final_price": 310000000.00,
    "collateral_unfrozen_for_others": true
  }
}
```

5. Event `NEW_LOT_PUBLISHED`

```
{
  "event": "NEW_LOT_PUBLISHED",
  "data": {
    "lot_id": "c1f76d22-192a-4c28-8742-1928374aef12",
    "title": "Hermes Birkin 25 Togo Gold Hardware",
    "brand": "Hermes",
    "category": "DESIGNER_BAGS",
    "starting_price": 250000000.00,
    "model_type": "CAPPED",
    "end_time": "2026-10-25T12:30:00.000Z"
  }
}
```

---

4. Dokumen Pengesahan (Document Sign-Off)

Spesifikasi kontrak antarmuka API (*API Contract Collection & Event Schemas*) ini telah ditinjau dan disetujui sebagai kontrak komunikasi baku antara tim Backend, Frontend Web (Next.js), dan Integrator Sistem.

PROJECT MANAGER **Geri Setyo Andrian**	TECH LEAD / TECHNICAL ARCHITECT **Alini**	SYSTEM ANALYST / SOLUTION ARCHITECT **Daniel H. S.**
		
Tanggal: 24 / 10 / 2026 **Status:** [✓] Disetujui [ ] Revisi	**Tanggal:** 24 / 10 / 2026 **Status:** [✓] Disetujui [ ] Revisi	**Tanggal:** 24 / 10 / 2026 **Status:** [✓] Disetujui [ ] Revisi